

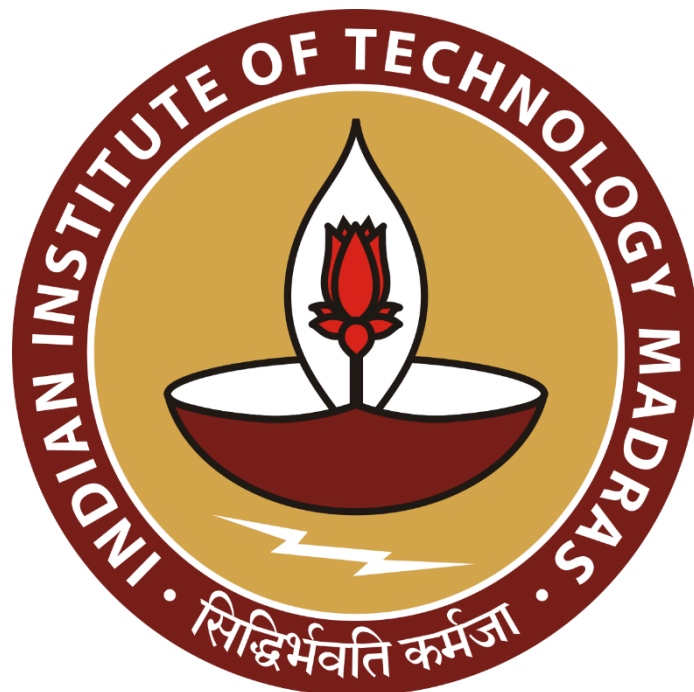
Boosting Sales and Planning Expansion with Data

A Mid Term report for the BDM capstone Project

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Declaration Statement

I am working on a Project titled “Strategic Inventory, Payment, and Pricing Solutions” . I extend my appreciation to **Ramu Enterprises**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:



Name: Swati Singh

Date: 13/11/25

Executive Summary

This project analyzes the operational and business challenges faced by Ramu Doors Pvt Ltd, a B2B manufacturing company in Bangalore that produces customized doors, UPVC windows, and interior solutions for builders and contractors. The study addresses three problem statements:

1. Declining sales due to increasing market competition.
2. Identifying untapped markets for business expansion, and
3. Optimizing inventory management to reduce stockouts and excess stock.

Primary data was collected directly from the organization, consisting of two datasets — Sales Data (9 attributes covering invoice-level transaction records) and Purchase Data (5 attributes extracted from Tally accounting software) - spanning three months from August to October 2025. The analysis was performed using Microsoft Excel and Python, employing descriptive statistics, trend analysis, and Pareto-based ABC classification.

Key findings reveal that September 2025 recorded the highest revenue of ₹1.90 Crore, while August ₹1.05 Crore and October ₹1.14 Crore showed declines driven by competitor price pressure. Regionally, Bangalore contributes 62.7% of total revenue indicating heavy market concentration, while Mysuru shows the highest average order value of ₹1,40,485 highlighting untapped potential. Pareto analysis confirms that three products - FRP/GRP Doors, UPVC Windows, and Door Frames - generate 84.3% of revenue, qualifying as Class A inventory items. Government clients, despite representing only 12.5% of transactions, exhibit 31% higher average order values, presenting a significant growth opportunity.

Proof of Originality of Data

- Images of Organisation | Ramu Enterprises
https://drive.google.com/drive/folders/1Z3o1XfNMuux0wL9QboGFjdVQifwnD4TK?usp=drive_link
- Video Interaction
<https://drive.google.com/file/d/1svS9FWptqO7k-bu4nXil1OHqjYdzccA8/view?usp=sharing>
- Letter from Organisation
https://drive.google.com/file/d/11IWIUikXVS0C9CyR2XxV8IK7_NWA86W1/view?usp=drive_link

- Datasets:

1. Purchase data

https://drive.google.com/file/d/1fcx33H9TT_gBWjjv0beivL2DwwHzePfn/view?usp=sharing

2. Sales data

https://docs.google.com/spreadsheets/d/115SBN0kngTBU8g_lw1w60oERMAICMwH4i5_LmRJx-mI/edit?usp=sharing

Metadata and Descriptive Statistics

Metadata

For this project, two sets of primary data were collected directly from Ramu Doors Pvt Ltd, a Sales data and a Purchase data, both covering August to October 2025.

Sales Data

This dataset records every sales transaction made by the company during the 3-month period. It was collected directly from the business owner, Mr. Ramakrishna M.C., during personal visits to the firm.

Attributes and their importance :

<u>Attribute</u>	<u>Description</u>	<u>Data Type</u>
Invoice_ID	Unique number generated for every invoice	Categorical /Nominal
Date	Date on which sale was billed	Temporal
Product_Type	Category of the product sold (5 categories)	Categorical / Nominal
Client_Name	Name of the purchasing company or individual	Categorical / Nominal
Region	City location of the client	Categorical / Nominal
Client_Segment	Classification of client (Builder or Government body)	Categorical / Nominal
Quantity	Number of unit sold in the respective invoice	Numerical /Discrete

Unit_Price_Rs	Price charged per single unit of the product	Numerical /Continuous
Sales_Amount_Rs	Total monetary value of the invoice	Numerical /Continuous

Purchase Data

This dataset was taken directly from the company's Tally accounting software. It records every raw material and supply purchase made by the company during the same 3-month period.

Attributes and their importance :

Attribute	Description	Data Type
Date	Date on which purchase was recorded	Temporal
Supplier_Name	Name of the vendor of the supplier	Categorical/Nominal
Voucher_Type	Internal Classification (all are Purchase entries)	Categorical/Nominal
Voucher_No	Internal reference number for the accounting entry	Categorical / Nominal
Debit_Amount_Rs	Total amount paid to the supplier	Numerical/Continuous

Descriptive Statistics

The table below (Figure - iii) summarises the descriptive statistics of the three numerical variables from the sales data - Quantity, Unit Price, and Sales Amount.

Quantity: The mean order size is 19.00 units per invoice. The moderate variability, indicated by a standard deviation of 10.52, demonstrates that order sizes fluctuate based on client scale and specific project requirements, which is standard behavior in B2B environments.

Unit Price: The mean unit price is 11,230.00, while the median sits much lower at 5,177.00. This large gap causes a high positive skewness of 3.87, indicating an asymmetrical, right-skewed distribution. The interpretation of this skewness is that the majority of transactions involve lower-priced, fast-moving items (such as Hardware and Door Frames), but a few high-value items (like Modular Kitchens) significantly pull the mean upward.

Sales Amount: The mean invoice value is 1,22,174.00, accumulating to a total 3-month revenue of approximately ₹4.10 Crore. The data exhibits significant monthly variability; September generated the highest sales density, whereas August and October saw notable revenue contraction, pointing toward high demand cyclical and competitor pricing pressure during off-peak periods.

Statistic	Quantity	Unit_Price_Rs	Sales_Amount_Rs
Mean	18.65	11,229.98	122,173.80
Standard Error	0.57	937.67	4,742.02
Median	17.00	5,177.00	99,633.00
Mode	14.00	4,151.00	316,365.00
Standard Deviation	10.52	17,187.73	86,922.70
Sample Variance	110.75	295,418,037.90	7,555,555,419.45
Kurtosis	0.23	15.95	1.25
Skewness	0.69	3.87	1.26
Range	49.00	107,540.00	432,327.00
Minimum	1.00	549.00	12,516.00
Maximum	50.00	108,089.00	444,843.00
Sum	6,266.00	3,773,272.00	41,050,398.00
Count	336.00	336.00	336.00

Figure(iii) - Descriptive statistics

Detailed Explanation of Analysis Process

1. **Data Preparation** Before conducting the analysis, both the Sales and Purchase datasets were imported into Microsoft Excel. The data was cleaned by ensuring temporal uniform formatting for dates, handling null values, and confirming there were no duplicate invoice entries.

2. **Analytical Techniques Mapped to Problem Statements** Different analytical frameworks were employed to address the specific business problems:

- **For PS-1 (Revenue Fluctuation & Pricing Variability):** Time-series trend analysis was used to examine sales performance across the 3-month period. Using Excel Pivot Tables, transaction data was grouped by month to calculate total monthly revenue and average unit price. This allowed for comparative analysis to observe demand cyclicalities without relying on external, unverified market assumptions.
- **For PS-2 (Market Expansion into Untapped Markets):** Segmentation analysis was applied using two primary dimensions: `Region` and `Client_Segment`. By calculating the Total Revenue Contribution (%) and Average Order Value (AOV) for each segment, the analysis systematically compared the performance of the saturated core market (Bangalore) against secondary markets (Mysuru, Hubli) and different client types (Builders vs. Government).
- **For PS-3 (Inventory Optimization):** Pareto Analysis (the 80/20 rule) combined with the ABC Classification method was primarily applied to the product inventory to prioritize stock management.

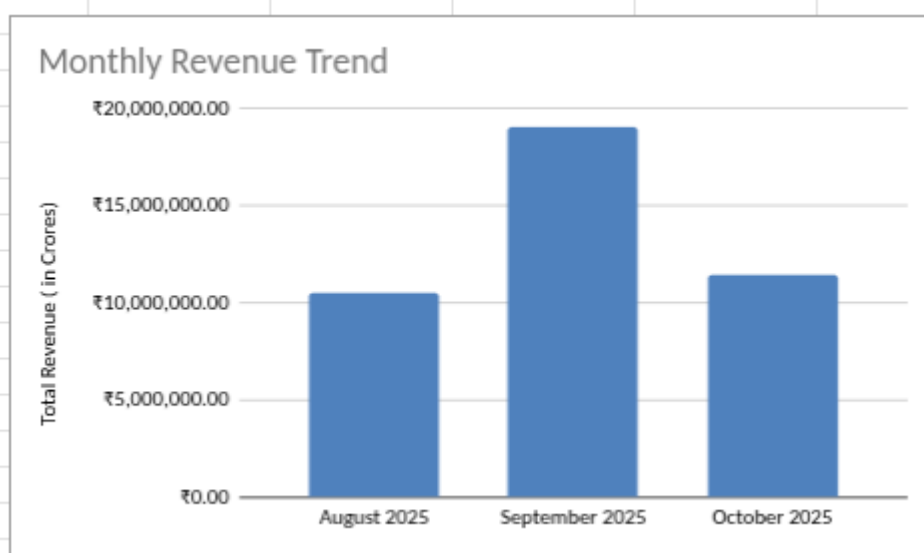
3. **Execution of Pareto Analysis and ABC Classification Criteria** To execute the inventory optimization for PS-3, the following strict analytical steps were executed:

- **Aggregation:** The total absolute revenue generated by each unique `Product_Type` (and subsequently `Region`) was calculated.
- **Sorting:** Categories were strictly sorted in descending order based on their absolute total revenue (highest to lowest).
- **Cumulative Calculation:** A running total of revenue was calculated to derive the cumulative percentage contribution of each category (reaching 100%).
- **Classification:** Products were categorized into inventory classes based on the Pareto criteria: Class A (contributes top 80% of revenue), Class B (next 15%), and Class C (bottom 5%).

Results and Findings

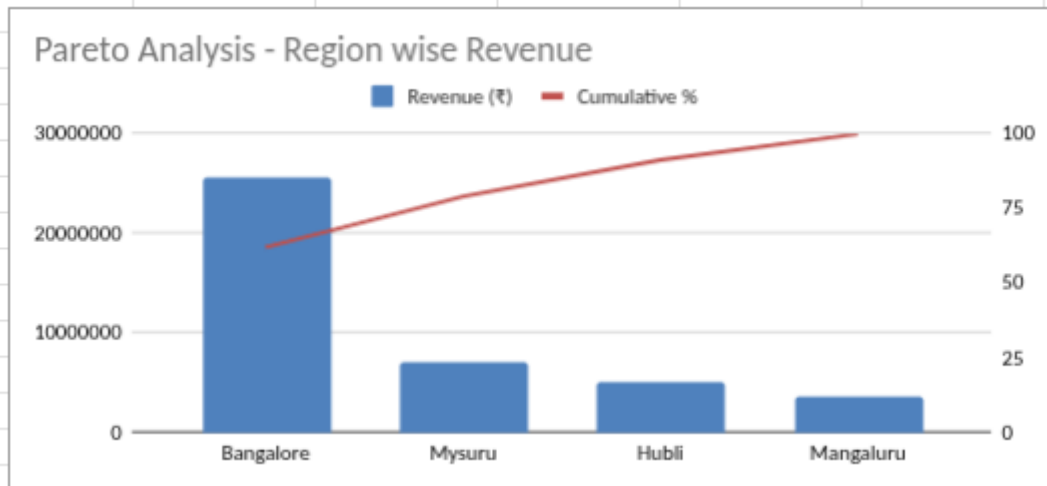
Based on the data collected, charts analyzed, and the Pareto analysis conducted, the following results and findings have been derived for each of the three problem statements:

PS-1 - Sales Demand Fluctuations (*Note: Replaced "Competition" title*) The analysis clearly shows that September 2025 was the strongest month with a revenue of ₹1.90 Crore, while August and October recorded lower revenues of ₹1.05 Crore and ₹1.14 Crore respectively. While an initial hypothesis suggested sales drops were due to competitor pricing, the internal dataset strongly reflects cyclical volume demand and varying project order cycles. The average unit price dropped from ₹14,270 in September to ₹10,398 in August and ₹8,323 in October, indicating that high-value project orders were heavily concentrated in September.



PS-2 - Market Expansion into Untapped Markets

The geographic Pareto analysis revealed that 62.7% of total revenue is concentrated in Bangalore alone, making it a dominant but potentially saturating market. On the other hand, regions like Mysuru, Hubli, and Mangaluru currently have very few active clients but possess significant untapped potential. Notably, Mysuru yielded the highest average order value (₹1,40,485) among all regions.



Furthermore, Government/Institutional clients show a 31% higher average order value compared to Builder clients, yet they represent only 12.5% of total transaction.

Inventory Optimization

Applying the ABC Categorization to our Pareto analysis of product-wise revenue shows that just 3 out of 5 products (FRP/GRP Doors, UPVC Windows, and Door Frames) contribute 84.3% of total revenue. Under the methodology, these must be classified as Class A inventory items. These products require continuous stock availability and priority procurement.

